Full dataset PCA

2024-11-06

Load gt

```
mad_bor_g1000_sum <-
  gt_load("../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld.gt")
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 2744
```

```
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

Count individuals from each group

```
malagasy_data <- subset(
  mad_bor_g1000_sum,
  mad_bor_g1000_sum$population %in% c(
    "Mikea", "Vezo",
    "Temoro", "Diego",
    "Merina",
    "Betsimisaraka",
    "Sakalava_Tsimihety",
    "Betsileo"
  )
)
nrow(malagasy_data)
```

```
## [1] 232
```

Impute

```
mad_bor_g1000_sum <- gt_impute_simple(mad_bor_g1000_sum, method = "mean")
gt_set_imputed(mad_bor_g1000_sum, TRUE)
gt_save(
  mad_bor_g1000_sum,
  "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld_imputed.gt"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld_imputed.gt

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld_imputed.gt')

## [1] "../data/intermediate/rbind_data/mad_bor_g1000_sum_neo_clean_ld_imputed.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum_n
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum_n
```

Compute PCA

```
random_pca <- gt_pca_autoSVD(mad_bor_g1000_sum)
```

```
## Discarding 1 variant with MAC < 10 or MAF < 0.02.
##
## Phase of clumping (on MAF) at r^2 > 0.2.. keep 190114 variants.
##
## Iteration 1:
## Computing SVD..

## The default of 'doScale' is FALSE now for stability;
##   set options(mc_doScale_quiet=TRUE) to suppress this (once per session) message

## 52 outlier variants detected..
## 2 long-range LD regions detected..
##
## Iteration 2:
## Computing SVD..
## 0 outlier variant detected..
##
## Converged!
```

```r
pcs <- as.data.frame(random_pca$u)
variance <- tidy(random_pca)
```

Add population metadata

```r
pop_names <-
  read.csv("../data/raw/data_input/igsr-1000-genomes-phase-3-release.tsv",
    sep = "\t"
  )

# find matching id in pop_names Sample.name and mad_bor_g1000_sum id
# for the matches, replace the population with the population in pop_names
for (i in seq_len(nrow(pop_names))) {
  if (pop_names$Sample.name[i] %in% mad_bor_g1000_sum$id) {
    mad_bor_g1000_sum[mad_bor_g1000_sum$id == pop_names$Sample.name[i], "population"] <- pop_names$Popul
  }
}

# replace "Luhya,Luhya" with "Luhya"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Luhya,Luhya"] <- "Luhya"
# replace "Esan,Esan" with "Esan"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Esan,Esan"] <- "Esan"
# replace "Gambian Mandinka,Gambian" with "Gambian Mandinka"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Gambian Mandinka,Gambian"] <- "Gambian Ma
# replace "Mende,Mende" with "Mende"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Mende,Mende"] <- "Mende"
# replace "Kinh,Kinh Vietnamese" with "Kinh Vietnamese"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Kinh,Kinh Vietnamese"] <- "Kinh Vietnames
# replcae "Punjabi,Punjabi" with "Punjabi"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Punjabi,Punjabi"] <- "Punjabi"
# replcae "Benga,Bengali" with "Bengali"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Bengali,Bengali"] <- "Bengali"

# replace any _ with a space
mad_bor_g1000_sum$population <- gsub("_", " ", mad_bor_g1000_sum$population)
```

Plot PCA

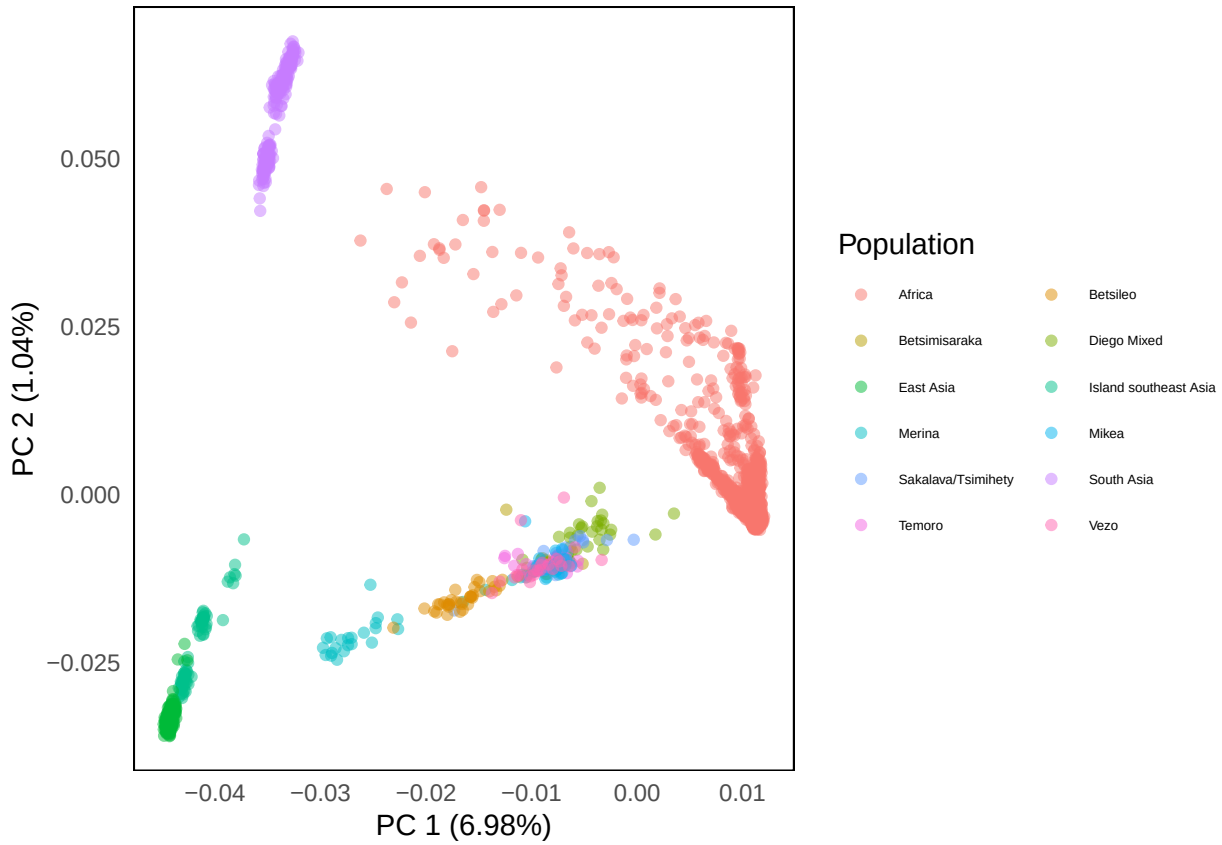
```r
# create a new factor column with Malagasy populations as 1 and others as 0
mad_bor_g1000_sum$malagasy <- 0
# rename mad_bor_g1000_sum$population "Sakalava Tsimihety" to "Sakalava/Tsimihety"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Sakalava Tsimihety"] <- "Sakalava/Tsimihe
# rename mad_bor_g1000_sum$population "Diego" to "Diego Mixed"
mad_bor_g1000_sum$population[mad_bor_g1000_sum$population == "Diego"] <- "Diego Mixed"
all_malagasy <- subset(
  mad_bor_g1000_sum,
  population %in% c(
    "Betsimisaraka", "Merina", "Diego Mixed",
```

```
    "Sakalava/Tsimihety", "Betsileo",
    "Vezo", "Mikea", "Temoro"
  )
)
mad_bor_g1000_sum[mad_bor_g1000_sum$id %in% all_malagasy$id, "malagasy"] <- 1

# create Region
mad_bor_g1000_sum$Region <- "Other"
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Borneo", "Sumba", "Timor"), "Region"] <- "Island
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Bengali", "Punjabi"), "Region"] <- "South Asia"
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Southern Han Chinese", "Kinh Vietnamese"), "Regi
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Esan", "Gambian Mandinka", "Luhya", "Mende", "Yo
mad_bor_g1000_sum[mad_bor_g1000_sum$malagasy == 1, "Region"] <- mad_bor_g1000_sum[mad_bor_g1000_sum$mala
# whareever is left as "Other" set to Africa
mad_bor_g1000_sum[mad_bor_g1000_sum$Region == "Other", "Region"] <- "Africa"



ggplot(pcs, aes(x = V1, y = V2, colour = mad_bor_g1000_sum$Region)) +
  geom_point(alpha = 0.5) +
  labs(
    title = NULL,
    x = paste0("PC 1 (", round(variance$percent[1], 2), "%)"),
    y = paste0("PC 2 (", round(variance$percent[2], 2), "%)"),
    colour = "Population"
  ) +
  guides(colour = guide_legend(ncol = 2, byrow = TRUE, size = 2)) +
  theme_minimal() +
  theme(
    legend.text = element_text(size = 5),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```



This PCA uses the KING relatedness filtering of 0.088, equivalent to second degree relatives, filtered within each population.

Unbalanced sample sizes will cause bias in the dimensions of this first plot. To explore how this might be affecting the PCA, the next dataset will randomly subsample populations with more individuals down to equivalent sizes.

Create a dataset with broadly equal sample sizes

```
set.seed(1)

subsample <- function(x) {
  if (nrow(x) <= 30) {
    return(x$id)
  } else {
    x <- sample_n(x, size = 30)
    return(x$id)
  }
}

keep <- mad_bor_g1000_sum %>%
  group_by(population) %>%
  group_map(~ {
    if (!(.y$population %in% c(
```

```
      "Betsimisaraka", "Merina", "Diego Mixed",
      "Sakalava/Tsimihety", "Betsileo",
      "Vezo", "Mikea", "Temoro"
    ))) {
      subsample(.x)
    } else {
      .x$id
    }
  })

mad_bor_g1000_sum <- mad_bor_g1000_sum %>% filter(id %in% unlist(keep))

if (file.exists("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_neo_ld_subsampled.gt")) {
  file.remove("../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_neo_ld_subsampled.gt")
}
```

```
## [1] TRUE
```

```
gt_save(
  mad_bor_g1000_sum,
  "../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_neo_ld_subsampled.gt"
)
```

```
##
## gen_tibble saved to ../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_neo_ld_subsampled.gt

## using FBM RDS: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g

## with FBM backing file: /home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/m

## make sure that you do NOT delete those files!

## to reload the gen_tibble in another session, use:

## gt_load('../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_neo_ld_subsampled.gt')

## [1] "../data/intermediate/rbind_data/mad_bor_g1000_sum_clean_neo_ld_subsampled.gt"
## [2] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum_m
## [3] "/home/EJC199/Dropbox/gitdata/malagasy_diversity/data/intermediate/rbind_data/mad_bor_g1000_sum_m
```

Loci and sample count

```
nrow(mad_bor_g1000_sum)
```

```
## [1] 1955
```

```r
count_loci(mad_bor_g1000_sum)
```

```
## [1] 198196
```

Count individuals from each group

```r
malagasy_data <- subset(
  mad_bor_g1000_sum,
  mad_bor_g1000_sum$population %in% c(
    "Mikea", "Vezo",
    "Temoro", "Diego Mixed",
    "Merina",
    "Betsimisaraka",
    "Sakalava_Tsimihety",
    "Betsileo"
  )
)

nrow(mad_bor_g1000_sum) - nrow(malagasy_data)
```

```
## [1] 1755
```

Compute PCA

```r
sub_pca <- gt_pca_randomSVD(mad_bor_g1000_sum)
pcs <- as.data.frame(sub_pca$u)
variance <- tidy(sub_pca)
```

Plot PCA

```r
# create a new factor column with Malagasy populations as 1 and others as 0
mad_bor_g1000_sum$malagasy <- 0
all_malagasy <- subset(
  mad_bor_g1000_sum,
  population %in% c(
    "Betsimisaraka", "Merina", "Diego Mixed",
    "Sakalava/Tsimihety", "Betsileo",
    "Vezo", "Mikea", "Temoro"
  )
)
mad_bor_g1000_sum[mad_bor_g1000_sum$id %in% all_malagasy$id, "malagasy"] <- 1
```

Plot PCA

```
# create a new factor column with Malagasy populations as 1 and others as 0
mad_bor_g1000_sum$mad_vs_global <- "Other"

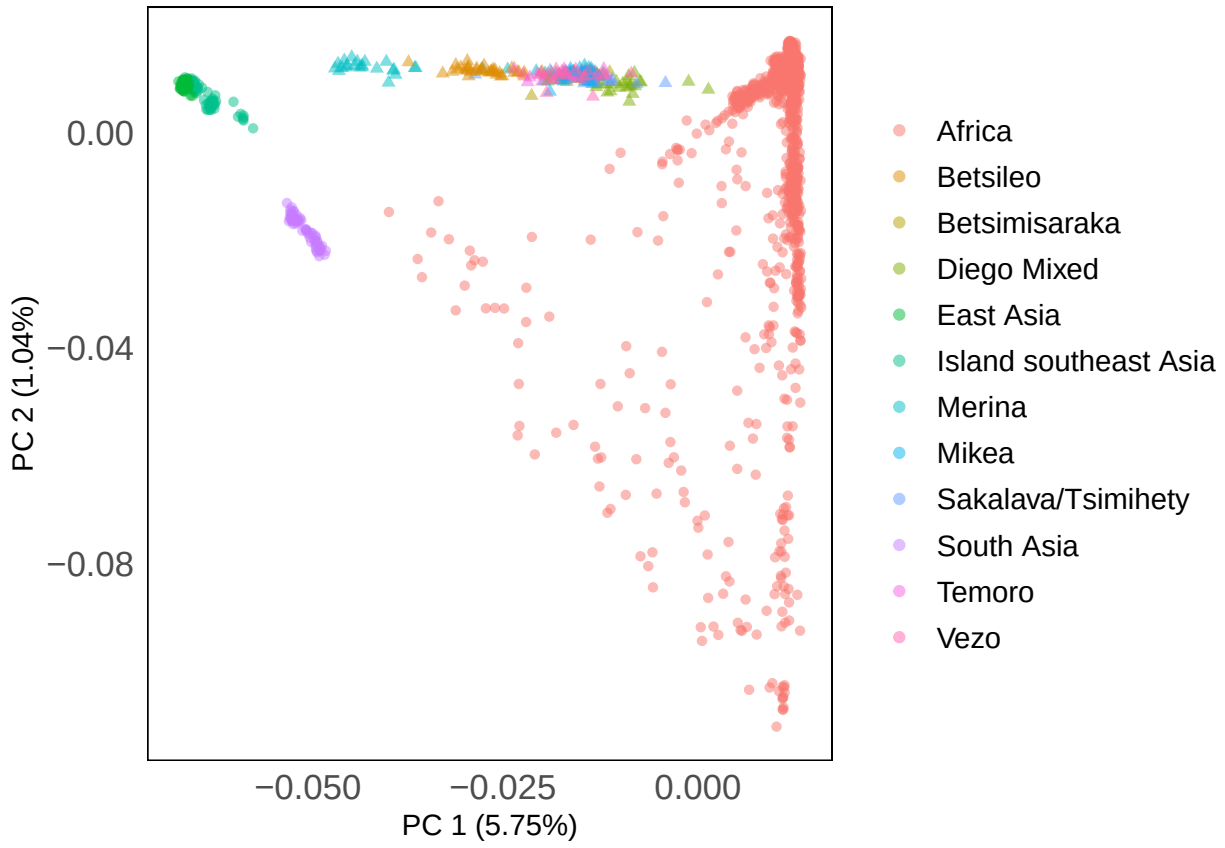
fill <- function(x) {
  if (x["malagasy"] == 1) {
    x["mad_vs_global"] <- "Malagasy"
  } else {
    x["mad_vs_global"] <- x["population"]
  }
}

mad_bor_g1000_sum$mad_vs_global <- apply(mad_bor_g1000_sum, 1, FUN = fill)
```

```
mad_bor_g1000_sum$Region <- "Other"

mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Borneo", "Sumba", "Timor"), "Region"] <- "Island
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Bengali", "Punjabi"), "Region"] <- "South Asia"
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Southern Han Chinese", "Kinh Vietnamese"), "Regi
mad_bor_g1000_sum[mad_bor_g1000_sum$population %in% c("Esan", "Gambian Mandinka", "Luhya", "Mende", "Yo
mad_bor_g1000_sum[mad_bor_g1000_sum$malagasy == 1, "Region"] <- mad_bor_g1000_sum[mad_bor_g1000_sum$mala
# whareryer is left as "Other" set to Africa
mad_bor_g1000_sum[mad_bor_g1000_sum$Region == "Other", "Region"] <- "Africa"


ggplot(pcs, aes(
  x = V1, y = V2,
  colour = mad_bor_g1000_sum$Region,
  shape = as.factor(mad_bor_g1000_sum$malagasy)
)) +
  geom_point(alpha = 0.5) +
  labs(
    title = NULL,
    x = paste0("PC 1 (", round(variance$percent[1], 2), "%)"),
    y = paste0("PC 2 (", round(variance$percent[2], 2), "%)"),
    colour = NULL, shape = "none"
  ) +
  theme(legend.position = "right") +
  guides(
    shape = "none"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(size = 16),
    axis.text = element_text(size = 13),
    legend.text = element_text(size = 11),
    axis.line = element_blank(),
    panel.grid.major = element_blank(),
    panel.grid.minor = element_blank(),
    panel.background = element_rect(fill = "white")
  )
```



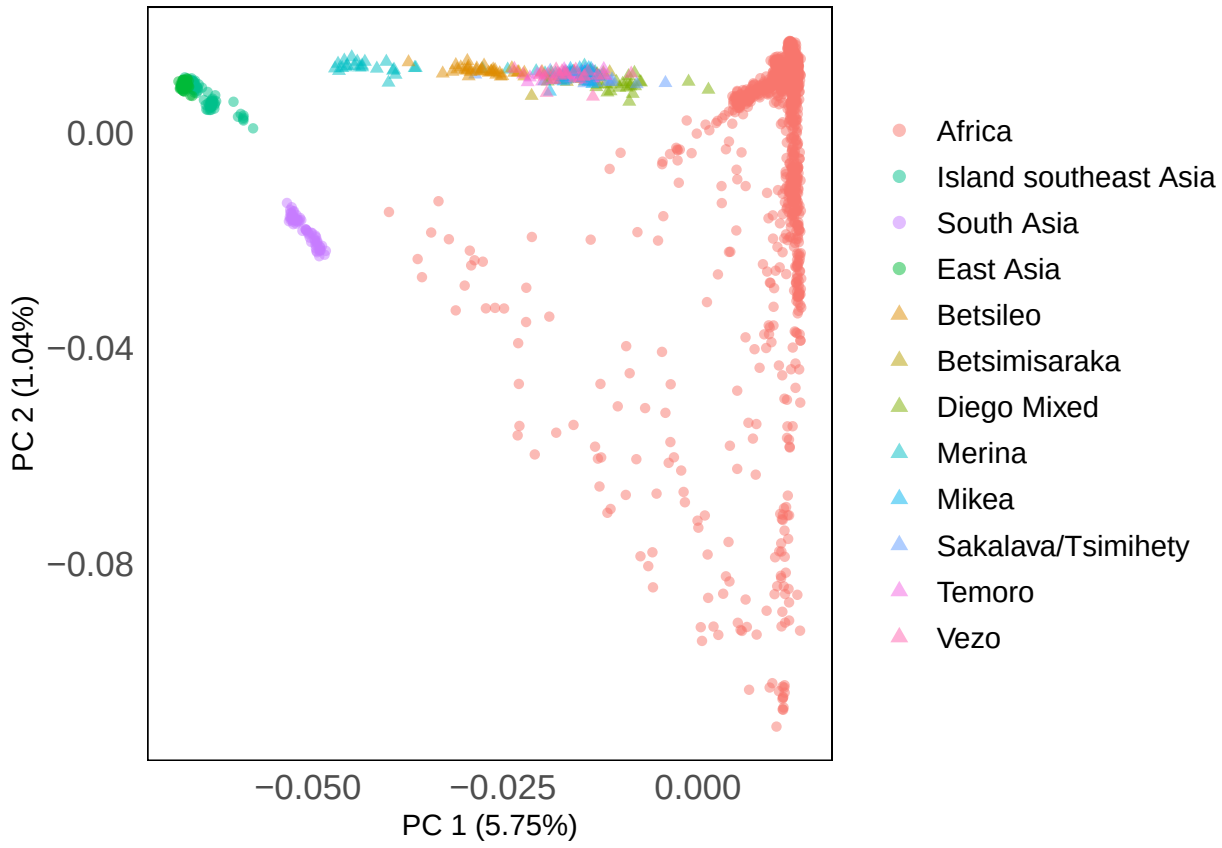

```
ggsave(paste0(res_dir, "pca_subsampled_regions_together.jpg"),
  width = 7, height = 7, dpi = 300
)
```

Altering the sample sizes reduces the explained variance of both principal components and increases the dispersion of the data.

```
mad_bor_g1000_sum$Region <-
  factor(mad_bor_g1000_sum$Region,
    levels = c(
      "Africa", "Island southeast Asia", "South Asia",
      "East Asia", "Betsileo", "Betsimisaraka", "Diego Mixed",
      "Merina", "Mikea", "Sakalava/Tsimihety", "Temoro", "Vezo"
    )
  )

ggplot(pcs, aes(
  x = V1, y = V2,
  colour = mad_bor_g1000_sum$Region,
  shape = as.factor(mad_bor_g1000_sum$malagasy)
)) +
  geom_point(alpha = 0.5) +
  labs(
    title = NULL,
    x = paste0("PC 1 (", round(variance$percent[1], 2), "%)"),
```

```
  y = paste0("PC 2 (", round(variance$percent[2], 2), "%)"),
  colour = NULL, shape = "none"
) +
theme(legend.position = "right") +
guides(
  shape = "none",
  colour = guide_legend(override.aes = list(
    shape = c(
      16, 16, 16, 16,
      17, 17, 17, 17,
      17, 17, 17, 17
    ),
    size = 2
  ))
) +
scale_colour_manual(
  values = c(
    "Africa" = "#F8766D",
    "East Asia" = "#00BA38",
    "South Asia" = "#C77CFF",
    "Island southeast Asia" = "#00C08B",
    "Betsileo" = "#DE8C00",
    "Betsimisaraka" = "#B79F00",
    "Diego Mixed" = "#7CAE00",
    "Merina" = "#00BFC4",
    "Mikea" = "#00B4F0",
    "Sakalava/Tsimihety" = "#619CFF",
    "Temoro" = "#F564E3",
    "Vezo" = "#FF64B0"
  )
) +
theme_minimal() +
theme(
  plot.title = element_text(size = 16),
  axis.text = element_text(size = 13),
  legend.text = element_text(size = 11),
  axis.line = element_blank(),
  panel.grid.major = element_blank(),
  panel.grid.minor = element_blank(),
  panel.background = element_rect(fill = "white")
)
```




```
ggsave(paste0(res_dir, "pca_subsampled_regions_together_large_legend.jpg"),
  width = 7, height = 5, dpi = 300
)
```

```
#Shape scale for malagasy is c(1, 3, 7, 15, 17, 19, 21, 23)
#Colour scale for malagasy is   scale_color_brewer(palette = "Dark2") + scale_fill_brewer(palette = "Da

# If we made all regional populations

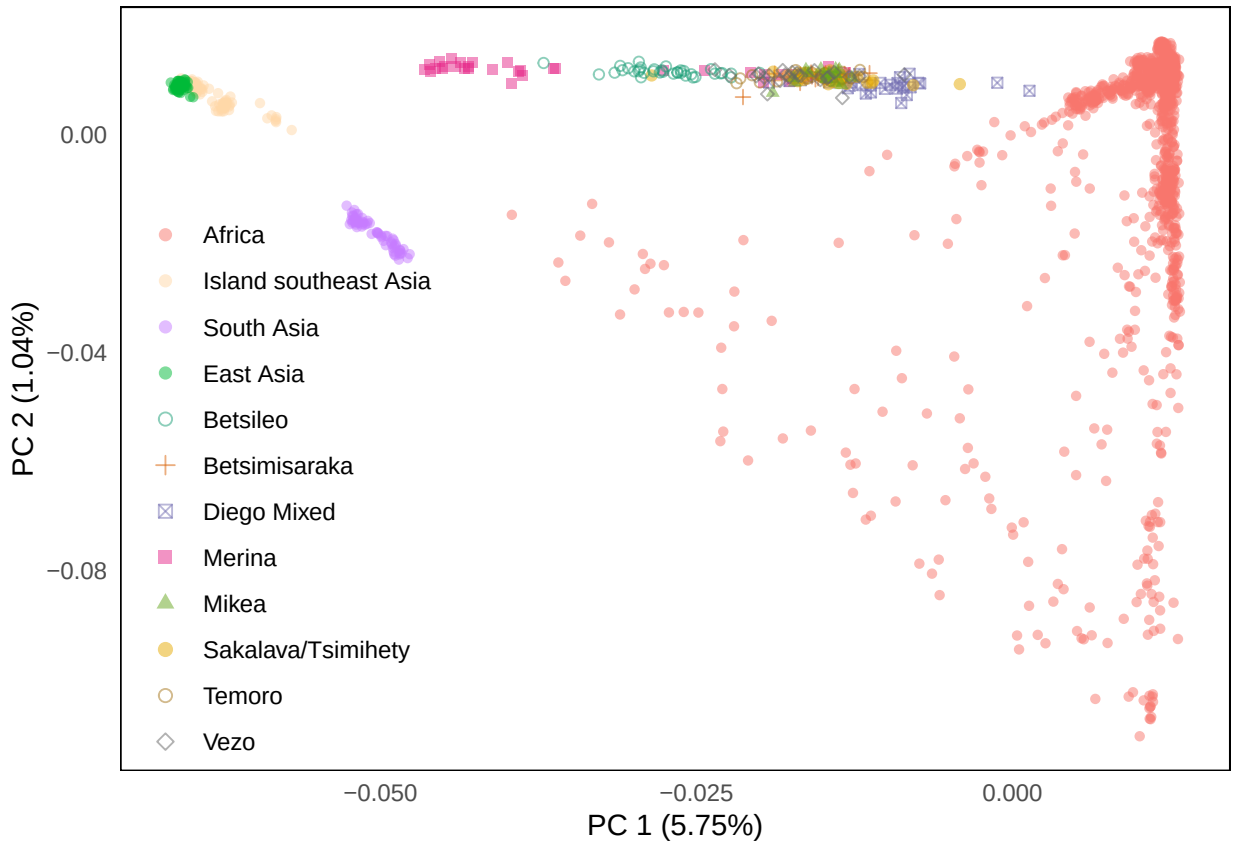
# if $malagasy is 1, then it becomes $population, otherwise it becomes "Other"
mad_bor_g1000_sum$shape <- ifelse(
  mad_bor_g1000_sum$malagasy == 1,
  mad_bor_g1000_sum$population,
  "Other"
)


p2 <- ggplot(pcs, aes(
  x = V1, y = V2,
  colour = mad_bor_g1000_sum$Region,
  shape = as.factor(mad_bor_g1000_sum$shape)
)) +
  geom_point(alpha = 0.5) +
  labs(
    title = NULL,
    x = paste0("PC 1 (", round(variance$percent[1], 2), "%)"),
```

```
  y = paste0("PC 2 (", round(variance$percent[2], 2), "%)"),
  colour = NULL, shape = "none"
) +
theme(legend.position = "right") +
guides(
  shape = "none",
  colour = guide_legend(override.aes = list(
    shape = c(
      16, 16, 16, 16,
      1, 3, 7, 15, 17, 19, 21, 23
    ),
    size = 2
  ))
) +
scale_colour_manual(
  values = c(
    "Africa" = "#F8766D",
    "East Asia" = "#00BA38",
    "South Asia" = "#C77CFF",
    "Island southeast Asia" = "#FFD8A8",
    "Betsileo" = "#1B9E77",
    "Betsimisaraka" = "#D95F02",
    "Diego Mixed" = "#7570B3",
    "Merina" = "#E7298A",
    "Mikea" = "#66A61E",
    "Sakalava/Tsimihety" = "#E6AB02",
    "Temoro" = "#A6761D",
    "Vezo" = "#666666"
  )
) +
scale_shape_manual(values = c(
  "Other" = 16,
  "Betsileo" = 1,
  "Betsimisaraka" = 3,
  "Diego Mixed" = 7,
  "Merina" = 15,
  "Mikea" = 17,
  "Sakalava/Tsimihety" = 19,
  "Temoro" = 21,
  "Vezo" = 23
)) +
theme_minimal() +
theme(
  plot.title = element_text(size = 16),
  #axis.text = element_text(size = 13),
  #legend.text = element_text(size = 11),
  axis.line = element_blank(),
  panel.grid.major = element_blank(),
  panel.grid.minor = element_blank(),
  panel.background = element_rect(fill = "white"),
  legend.position = c(0.15, 0.37)
)
```

```
## Warning: A numeric ‘legend.position‘ argument in ‘theme()‘ was deprecated in ggplot2
## 3.5.0.
## i Please use the ‘legend.position.inside‘ argument of ‘theme()‘ instead.
## This warning is displayed once per session.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```

```
p2
```




```
ggsave(paste0(res_dir, "test_pca.jpg"),
  width = 7, height = 5, dpi = 300
)

saveRDS(p2, paste0(res_dir, "global_pca.rds"))
```